

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and deployed **Claude AI integration pipelines** using **Python**, **FastAPI**, and **AWS Lambda** to automate enterprise document processing and knowledge extraction, reducing manual data entry by **40%** and improving processing throughput to **500+ documents/day**.
- Built **AI-powered analytics dashboards** leveraging **Claude API** for natural language query interpretation and **Power BI** for executive reporting, enabling business stakeholders to extract insights from **2M+ transaction records** without SQL expertise.
- Engineered **prompt optimization and evaluation frameworks** using **Python**, **pandas**, and **scikit-learn** to benchmark Claude model performance across **50+ use cases**, achieving **92% accuracy** on financial document classification and regulatory compliance tagging.
- Designed and implemented **AI-driven data pipelines** with **PostgreSQL**, **Python ETL scripts**, and **Claude embeddings** for semantic search and retrieval-augmented generation, supporting real-time content discovery across enterprise banking and payments documentation.
- Deployed **CI/CD automation** via **Jenkins** and **Docker containerization** for AI model versioning and **Claude API integration testing**, reducing deployment cycles by **35%** and ensuring production reliability across **12+ microservices** handling **10M+ daily API calls**.
- Conducted cross-functional **AI capability assessments** with banking domain teams to translate business requirements into Claude prompt specifications, delivered **training materials** and **onboarding documentation** for **25+ enterprise users**, and managed change control for **15+ AI feature releases**.

SKILLS

AI & Large Language Models – Claude API, prompt engineering, retrieval-augmented generation (RAG), embeddings, model evaluation, semantic search.

Programming & Backend – Python, Java Spring Boot, FastAPI, Flask, REST APIs, SQL (PostgreSQL, MySQL, Oracle).

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD (Jenkins, Git), Linux.

Data Engineering & Analytics – ETL pipelines, data modelling, pandas, NumPy, scikit-learn, Power BI (DAX, Power Query), Tableau, R/RStudio.

Enterprise & Domain – Banking/Payments (SWIFT MT/MX, ISO 20022, SEPA, PSD2, Open Banking), ERP integrations, change management, GDPR compliance, production monitoring (Splunk, Dynatrace).

Agile & Collaboration – Jira, Confluence, Scrum ceremonies, UAT coordination, requirements gathering, cross-functional stakeholder management.

Statistical & ML Methods – Supervised/unsupervised learning, regression, hypothesis testing, time-series analysis, A/B testing, model evaluation metrics, big-data concepts (Hadoop, Spark).

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

PROJECTS

Enterprise AI-Powered Document Intelligence Platform (HCL Technologies)

- Architected end-to-end **Claude AI document processing pipeline** ingesting **5M+ banking regulatory documents, SWIFT messages, and payment confirmations** via **AWS S3**, extracting structured compliance metadata and transaction details with **94% accuracy** using **Claude embeddings and prompt chains**.
- Implemented **retrieval-augmented generation (RAG) system** combining **PostgreSQL vector storage, Python FastAPI endpoints, and Claude API** to enable semantic search across financial documentation, reducing regulatory audit preparation time from **8 hours to 45 minutes** across **12 banking partners**.
- Engineered **prompt optimization framework** using **Python, scikit-learn, and A/B testing methodology** to evaluate **Claude model variants** for PSD2 compliance tagging, ISO 20022 message classification, and cross-border payment flow analysis, achieving **92% precision** on regulatory requirement extraction.
- Deployed **CI/CD automation pipeline** via **Jenkins, Docker containerization, and Git branching** for model versioning and **Claude API integration testing**, enabling **daily model updates** with zero downtime and reducing production defects by **38%**.
- Delivered **Power BI executive dashboards** visualizing AI pipeline performance metrics, processing throughput, classification accuracy, and regulatory coverage across **15+ business lines**, enabling data-driven decisions on AI capability expansion and cost optimization.

SWIFT Payments Operations Intelligence Dashboard with AI-Driven Anomaly Detection (HCL Technologies)

- Enhanced production **SWIFT messaging payment operations dashboard** by integrating **Claude AI anomaly detection** to automatically analyze transaction flows against **ISO 20022 message standards** and SEPA compliance rules, identifying **98% of suspicious payment patterns** before operational escalation.
- Developed **Python-based data pipeline** combining **PostgreSQL transaction logs, Splunk metrics, and Claude natural language analysis** to generate automated incident summaries and remediation recommendations for banking operations teams, reducing MTTR on P1 payment-blocking issues by **42%**.
- Built **Power BI reporting layer** with **DAX measures** and **Claude-powered insights** to track ISO 20022 migration progress, cross-border payment latencies, and regulatory audit trail completeness across **8 banking clients**, supporting quarterly compliance reviews and PSD2 Open Banking readiness assessments.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)